Joint Learning of Safety-Relevant Anatomy from Heterogeneous Supervision in Laparoscopic Cholecystectomy

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Abstract

Purpose: Preventing bile duct injury during laparoscopic cholecystectomy relies on recognizing safety-relevant anatomical structures and identifying regions where dissection should be avoided. We investigate how these complementary tasks can be learned jointly from heterogeneous expert annotations covering different anatomical targets and surgical conditions. **Methods:** We combine two new indocyanine green (ICG)-guided datasets with an extended white-light dataset and adapt SAM3 to box, mask, point and line supervision. We compare joint supervision of six anatomical targets with CBD-only training and

investigate sampling, phase, local fluorescence evidence and training-source composition. **Results:** The full joint model achieves NGZ mask AP50-95 of 58.89% and 52.36% on the two ICG datasets. In matched-budget experiments on a single ICG dataset, joint supervision improves CBD AP50 over CBD-only training by 3.44 and 5.18 points on the two ICG test sets. For cases with weak local fluorescence evidence, adding Endoscapes increases AP50 from 22.70% to 24.87%. **Conclusion:** Joint learning from complementary annotations generally improves CBD localization. Our controlled experiments reveal how frame sampling, surgical phase, local fluorescence evidence and training-data composition influence reported performance and cross-center generalization, highlighting important pitfalls and informing more reliable evaluation practices. Precise landmark recovery and CBD detection without local fluorescence remain challenging.

Keywords: laparoscopic cholecystectomy, no-go zone, anatomical landmarks, heterogeneous supervision, fluorescence imaging

1 Introduction

Bile duct injury (BDI) is a serious complication of laparoscopic cholecystectomy (LC), commonly associated with mistaking the common bile duct (CBD) or common hepatic duct for the cystic duct (CD) [5]. Preventing such errors requires anatomical orientation throughout dissection. The B-SAFE landmarks provide reference points for this purpose: Rouviere's sulcus (RS) and the base of liver segment IVb (S4b) help identify the plane of the CBD and delimit the region where dissection should be avoided [5, 15]. Together, they define the R4U safety line, above which dissection is recommended [5]. This clinical reasoning motivates recognizing related structures together, with delineation of the no-go zone (NGZ) as the principal safety-related target (Fig. 1).

Computer-vision approaches to safety in LC include critical view of safety (CVS) assessment [10, 11] and segmentation of safe and unsafe dissection regions [8]. Complementary studies detect anatomical landmarks [4, 16], construct guided dissection lines [14], or localize bile ducts in fluorescence-guided procedures [17]. Surgical-scene datasets also provide gallbladder (GB) and CD segmentation [11]. These studies address overlapping subsets of the anatomy relevant to safe dissection. Joint supervision of ducts, contextual anatomy, landmark lines and NGZ offers a way to bring these complementary targets into one model.

Beyond developing accurate recognition models, reliable evaluation is essential for assessing their potential clinical utility. Recent work has highlighted how inappropriate validation metrics, incomplete annotations and the temporal structure of surgical videos can lead to misleading performance estimates [9, 13]. These concerns are particularly relevant to laparoscopic cholecystectomy, where anatomical visibility and surgical conditions evolve throughout the procedure. Consequently, the composition of both training and evaluation datasets may substantially influence reported performance, potentially obscuring important limitations in safety-relevant anatomical recognition.

Obtaining comprehensive expert annotations for every target across diverse surgical views is, however, demanding. Experts must choose both what to annotate and when. Expert-selected keyframes emphasize clinically relevant views but may miss intervening conditions, whereas regular sampling broadens temporal coverage at greater annotation cost. As anatomical exposure varies across surgical phases, these choices affect both training supervision and measured performance.

Indocyanine green (ICG) fluorescence adds another source of variation. It can assist biliary identification [3], but administration protocols affect visualization [2], and visible fluorescence at the CBD location varies within a procedure [17]. Pooling ICG and white-light data from multiple centers therefore combines differences in imaging, local anatomical evidence and annotation coverage. Learning from such data requires distinguishing unavailable labels from reviewed negative targets [7].

We make two contributions. First, we introduce two new ICG-guided datasets for safety-relevant anatomical recognition: iBSAFE, acquired at a single center, and iBSAFE-MC, acquired at two additional centers. We additionally extend Endoscapes with expert-annotated CBD boxes. We adapt SAM3 to jointly learn from their complementary box, mask, point and line annotations. While previous safety-oriented approaches typically focus on narrower sets of anatomical targets, our framework jointly addresses six classes: GB, CD, CBD, RS, S4b and NGZ. This setup combines variation in centers, sampling, phases, fluorescence and supervised targets, with NGZ delineation accompanied by localization of the relevant anatomy. Second, through controlled experiments and stratified evaluations, we demonstrate how frame sampling, surgical phase, local fluorescence evidence and training-data composition influence reported performance. These findings reveal important pitfalls in evaluating safety-relevant anatomical recognition and provide practical recommendations for more reliable comparisons across surgical conditions and centers.

2 Materials and methods

2.1 Data description

Our principal target is the NGZ, annotated as a mask of the region where dissection should be avoided [8]. Its delineation is motivated by BDI prevention; the accompanying anatomical targets provide complementary information about the surgical scene. CBD and CD localization address the distinction central to duct misidentification. RS and S4b lines describe landmarks used for orientation within B-SAFE and for identifying the R4U safety boundary [5]. GB provides a comparatively conspicuous target and context for the geometry of the operative field. We additionally annotate local CBD fluorescence evidence: experts inspect the CBD box and, when fluorescence is visible within it, mark the most visible point. We call such cases "strong evidence", and "weak evidence" otherwise. This point supports supervision and analysis of detection difficulty according to local fluorescence evidence; it is not an annotation-confidence score. Its absence does not establish that the CBD is absent. Thus, we consider six structures or regions (GB, CD, CBD, RS, S4b and NGZ), with CBD boxes and fluorescence points serving as distinct annotation targets. Fig. 1 illustrates the complementary annotations used for joint learning.

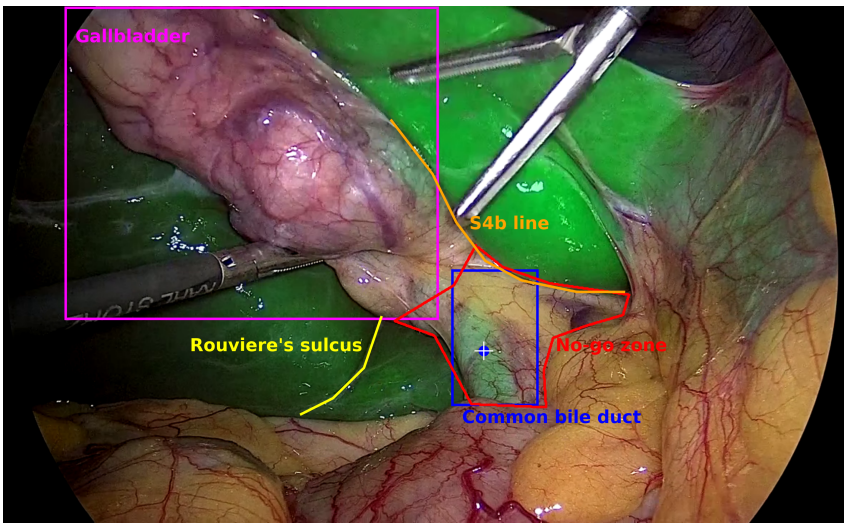


Fig. 1 Safety-relevant annotations on an iBSAFE frame: GB (magenta) and CBD (blue) boxes, RS (yellow) and S4b (orange) lines, and the NGZ boundary (red). The blue point marks local CBD fluorescence; the NGZ indicates where dissection should be avoided. CD is not annotated in this frame since it is not visible in the pre-HCT phase.

We distinguish two periods relevant to safety assessment. Pre-HCT precedes hepatocystic-triangle dissection, when orientation relies on the anatomy available before tissue clearance. HCT extends from dissection onset to cystic-duct transection, encompassing the exposure needed to identify structures before division [5]. Dissection can improve duct visibility, while changes in exposure and retraction may alter the visibility of surrounding landmarks. We therefore expect phase-related detection difficulty to depend on the structure. The shorter pre-HCT period permits denser regular annotation: iBSAFE-u and iBSAFE-MC are sampled every five seconds in pre-HCT and every 30 seconds in HCT. This allocation balances temporal coverage against annotation workload.

Tables 1 and 2 summarize the frame inventory and exhaustive annotations; Fig. 2 shows source-specific examples. The new iBSAFE cohort contains 123 videos from one center, with expert-selected keyframes (iBSAFE-k) and regularly sampled frames (iBSAFE-u) covering both phases. The 108 development videos comprise 86 training and 22 validation videos, with the validation set shared by iBSAFE-k/u; 15 videos are reserved for testing. All procedures used the same camera type (Storz 4K) and ICG protocol: intravenous injection 20–30 minutes before surgery. Two senior surgeons annotated iBSAFE through iterative refinement until agreement was reached. The keyframe and regular subsets provide complementary annotation types; S4b lines are available only in pre-HCT.

The new iBSAFE-MC dataset contains 41 procedures from two other centers: 8 training, 2 validation and 31 test videos, sampled at the same phase-specific intervals as iBSAFE-u. It includes multiple camera systems and ICG administration protocols selected according to surgeon preference. Administration differences may affect fluorescence intensity in the GB, liver and CBD. Each video was annotated by one senior

Table 1 Total frame inventory. iBSAFE-k/u share a cohort. iBSAFE-MC has one additional test frame with unknown phase, included in its total.

Subset	ICG	Sampling	Split	Videos	pre-HCT	HCT	Total
iBSAFE-k	Yes	Keyframes	Train+Val	108	323	3,087	3,410
			Test	15	59	405	464
iBSAFE-u	Yes	Uniform*	Train+Val	108	1,190	3,705	4,895
			Test	15	199	417	616
Endoscapes	No	Uniform, 30 s	Train	120	0	1,176	1,176
			Val	41	0	367	367
			Test	40	0	338	338
iBSAFE-MC	Yes	Uniform*	Train+Val	10	111	350	461
			Test	31	336	811	1,148

* Every 5 s in pre-HCT and 30 s in HCT.

Table 2 Exhaustive annotations for the six anatomical structures. CBD boxes and fluorescence points represent distinct targets. A dash indicates no exhaustive annotation used in that cell. S4b lines (*) are available in pre-HCT only.

Subset	GB	RS	S4b	CD	CBD	NGZ
iBSAFE-k	Box	Box	—	Box	Box	Mask
iBSAFE-u train	—	Line	Line*	—	Box/point	—
iBSAFE-u test	Box	Line/box	Line*	Box	Box/point	Mask
iBSAFE-MC	Box	Box	—	Box	Box	Mask
Endoscapes	Mask	—	—	Mask	Box	—

surgeon. Endoscapes comprises 201 non-ICG videos with an official split of 120 training, 41 validation and 40 test videos, sampled every 30 seconds during HCT [11]; we use its GB/CD masks and add CBD boxes. Together, these collections combine variation in centers, imaging, sampling, phases and target coverage rather than requiring every frame to carry every annotation type.

Missing annotations and reviewed negatives have different meanings. Where a target is exhaustively annotated, a reviewed frame without that target supplies negative supervision; absent CBD boxes in iBSAFE-u are such reviewed negatives. Where annotation is unavailable, as for RS in Endoscapes, the target remains unknown and is excluded from supervision. This distinction concerns the annotation target in the image, not anatomical absence from the patient. All splits are disjoint at video level: every frame from a video belongs to the same split, including across iBSAFE-k/u. Optimization counts exclude fixed validation cases and reflect experiment-specific subset selection.

2.2 Joint training

The full joint model uses all available optimization data from the three collections, combining iBSAFE keyframes and regular frames. It learns NGZ delineation alongside anatomical localization from complementary supervision, without requiring complete annotations in each image. SAM3 associates fixed anatomical text prompts with image

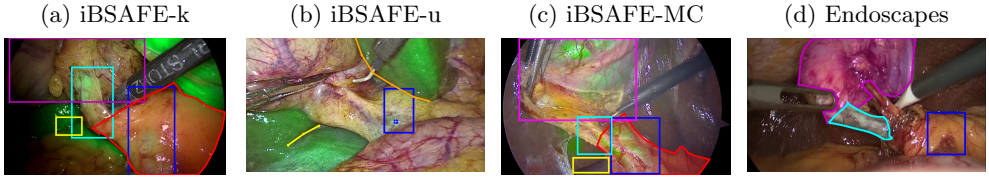


Fig. 2 Examples of the complementary views and annotations used for joint anatomical localization. Colors identify GB (magenta), CD (cyan), CBD (blue), RS (yellow), S4b (orange) and NGZ (red). The blue point in (b) marks local fluorescence evidence at the CBD location.

regions [1]. We adapt the model with LoRA while freezing its pretrained parameters, optimizing approximately 0.6% of the model [6].

For each image, losses are applied only to supported annotation types and targets. Reviewed negatives contribute background supervision, whereas missing labels are ignored rather than treated as background. Available positive annotations contribute the corresponding spatial losses, giving

$$\mathcal{L} = \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{mask}} + \lambda_p \mathcal{L}_{\text{point}} + \lambda_l \mathcal{L}_{\text{line}}. \tag{1}$$

Box supervision combines classification, L1 and generalized-IoU losses, while mask supervision combines focal and Dice losses. We set $\lambda_p = 1$ and $\lambda_l = 5$. Point supervision encourages foreground predictions at annotated locations without treating missing points as negatives. For line supervision, we sample 128 points uniformly along each annotated curve and encourage foreground predictions along the curve and background outside a two-pixel tolerance band. Curves also provide enclosing boxes, and point/line losses are averaged over contributing annotations. Detailed loss formulations are provided in Supplementary Methods S1. This treatment of partial annotations enables joint learning from complementary datasets [7]. Inference uses images and class names, without phase or evidence labels.

All-target models select checkpoints by validation NGZ mask AP50; CBD-only models use CBD box AP50. For the full joint model, NGZ AP50 is averaged without weighting across iBSAFE and iBSAFE-MC validation sets; Endoscapes has no NGZ targets, and official tests are excluded from selection. To balance exposure to sources of different sizes, sources are sampled in proportion to the square root of their unique optimization-frame counts, with weights recomputed when sources are omitted. This increases the relative contribution of smaller sources compared with sampling all frames uniformly, while retaining greater exposure to larger sources. Implementation settings and source-sampling details are provided in Supplementary Methods S1–S2.

2.3 Metrics

Following recent recommendations on reliable medical image analysis validation [9, 13], we distinguish evaluation populations according to sampling strategy, surgical phase, local fluorescence evidence and surgical center. We use matched test populations for model comparisons and stratified analyses to characterize performance differences that aggregate metrics may conceal.

Table 3 Full joint-model performance on official test sets (%). iBSAFE box/mask results use deduplicated keyframe and uniform test frames; line F1@5 uses 616 RS and 199 pre-HCT S4b uniform frames. Dashes indicate unavailable annotations.

Target	Endpoint	iBSAFE	iBSAFE-MC	Endoscapes
NGZ	Mask AP50–95	58.89	52.36	—
GB	Box AP50	93.39	98.12	84.62
CD	Box AP50	64.29	51.39	62.96
CBD	Box AP50	80.98	79.41	50.97
RS	Box AP50	33.51	39.39	—
RS	Line F1@5	10.23	—	—
S4b	Line F1@5	27.65	—	—

Only annotation-supported classes and frames are scored. Predictions are matched to same-class references in descending confidence order, with each reference matched at most once. Unmatched predictions are false positives and unmatched references are false negatives. AP50 uses intersection-over-union threshold 0.5; AP50–95 averages AP over ten thresholds from 0.50 to 0.95. Detection uses box overlap and NGZ segmentation uses mask overlap.

For RS/S4b, mask logits are mapped to native cropped-image coordinates and binarized at > 0. Class-wise box NMS uses IoU 0.5 and retains at most 100 detections; detections with confidence ≥ 0.5 contribute to each class's mask union. Zhang–Suen skeletonization yields a thin curve, without component selection or branch pruning. Reference polylines are rasterized at one-pixel width. Curve precision and recall measure the fractions of predicted and reference pixels within tolerance δ of the other curve; their harmonic mean gives $F1_\delta$ [12]. Matched pixel counts are accumulated across eligible frames. Native-grid distances are scaled to a longest edge of 1008 pixels, with $\delta = 5$ for primary results. This evaluation tolerance is distinct from the training band. Further evaluation details appear in Supplementary Methods S3.

3 Experiments

3.1 Joint learning of safety-relevant targets

We first evaluate the full joint SAM3 model trained on all three collections, using NGZ segmentation as the principal target and anatomical localization as complementary endpoints (Table 3). NGZ mask AP50–95 reaches 58.89% on iBSAFE and 52.36% on iBSAFE-MC. CBD box AP50 reaches 80.98%, 79.41% and 50.97% on iBSAFE, iBSAFE-MC and Endoscapes, respectively. The lower Endoscapes score reflects the beneficial impact of ICG evidence in the other datasets. GB localization is comparatively strong, whereas RS/S4b line F1@5 remains low (10.23%/27.65% on iBSAFE). Thus, joint learning supports complementary targets, but approximate localization does not ensure precise landmark recovery. Figure 3 illustrates predictions and failure cases.

To compare joint and single-target training for CBD localization, we compare iBSAFE-trained joint and CBD-only models under keyframe, uniform and combined training policies (Table 4). Unlike the full model above, these models use iBSAFE

Table 4 CBD box AP50 (%) under joint and CBD-only training. All models use matched training budgets and identical test sets. Each configuration is evaluated in a single run.

	iBSAFE		iBSAFE-MC	
Training policy	Joint training	CBD-only	Joint training	CBD-only
Keyframes	**76.29**	75.17	**69.99**	68.67
Uniform	**82.20**	78.43	70.89	**71.78**
Combined	**82.01**	78.57	**78.15**	72.97

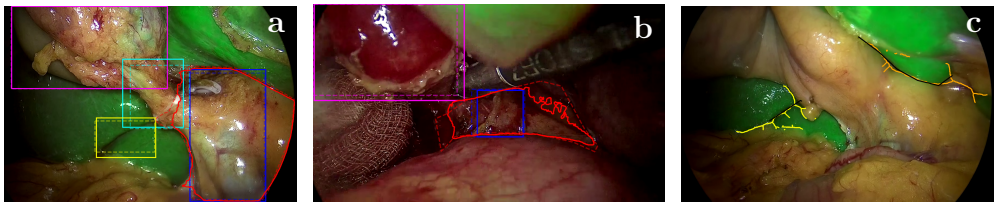


Fig. 3 Qualitative examples from the full joint model. (a) Joint anatomical localization. (b) CBD localization with limited local fluorescence evidence. (c) Joint RS/S4b line predictions, showing alignment but extra branches. Solid colored boxes and region contours denote predictions; dashed contours denote references. RS and S4b predictions are in yellow and orange respectively, while their ground-truth references are both in black. Other class colors follow Fig. 2.

training data only. All six runs use 2,162 training frames and complete 12,000 optimizer updates; paired models use identical frames and the checkpoint criteria described in Methods. All policies are evaluated on the same combined iBSAFE test population and the same external iBSAFE-MC test set. In five of the six comparisons, joint training improves CBD box AP50 over CBD-only training, with gains of 1.12–3.77 points on iBSAFE and 1.32–5.18 points on iBSAFE-MC. The only exception is uniform training on iBSAFE-MC, where joint supervision decreases AP50 by 0.89 points.

3.2 Impact of sampling and surgical phase on evaluation

We stratify the fixed full joint model's iBSAFE-u results by surgical phase (Table 5). NGZ mask AP50-95 increases from 38.87% in pre-HCT to 57.22% in HCT, an 18.35-point difference. In contrast, CBD AP50 remains nearly unchanged, while GB localization performs better in pre-HCT. These findings demonstrate that phase composition can substantially influence aggregate performance estimates and conceal phase-specific limitations, motivating stratified reporting or comparisons on matched phase distributions.

Frame sampling introduces another source of evaluation variability. In matched-budget experiments, evaluating identical models on expert-selected keyframes rather than regularly sampled frames increases mean CBD AP50 by 3.4-6.7 points, depending on the training policy (Supplementary Section S4). Uniform training also improves uniform-test AP50 by 4.4 points over keyframe training. These results highlight the importance of reporting sampling strategies and comparing methods on common evaluation populations.

Table 5 Phase-stratified results on iBSAFE-u (199 pre-HCT and 417 HCT frames; %). CD AP is undefined in pre-HCT; S4b has no HCT line references. RS line F1 includes reviewed negatives (83/12 positive pre-HCT/HCT frames).

Target	Endpoint	pre-HCT	HCT
GB	Box AP50	94.43	90.16
CBD	Box AP50	78.11	78.41
CD	Box AP50	—	60.06
RS	Box AP50	22.06	29.73
RS	Line F1@5	13.84	0.00
S4b	Line F1@5	27.65	—
NGZ	Mask AP50–95	38.87	57.22

Table 6 CBD localization on the full iBSAFE-u test set (both phases). B: iBSAFE; E: Endoscapes. The absent-point group has 271 frames and 152 CBD boxes; the present-point group has 345 frames and 345 boxes. Scores are percentages.

CBD fluorescence evidence	Training data	AP50
Weak	B	22.70
	B + E	24.87
Strong	B	94.13
	B + E	94.27

3.3 Influence of local fluorescence evidence on performance assessment

We compare iBSAFE-only and iBSAFE+Endoscapes joint models on all 616 iBSAFE-u test frames across both phases (Table 6), matching training recipe apart from source inclusion and reweighting. Both models use all training frames for 12,000 updates and identical test groups; prevalence, phase and exposure may contribute to between-group differences.

On weak evidence samples, Endoscapes adds 2.17 AP50 points, while on strong evidence samples, AP50 changes little. These results show that CBD localization performance varies markedly with local fluorescence evidence, with AP50 remaining particularly low on weak-evidence frames. Although adding white-light training data provides a modest improvement in this setting, the large performance gap between fluorescence strata illustrates how aggregate scores may conceal challenging imaging conditions. Since the strata also differ in CBD prevalence and potentially surgical phase, this gap cannot be attributed solely to fluorescence visibility. Reporting performance separately according to local fluorescence evidence therefore provides a more informative assessment of CBD detection robustness.

3.4 Cross-center generalization and training-data composition

We compare joint SAM3 training-source combinations on the same official iBSAFE-MC test set (Table 7). Models excluding iBSAFE-MC from both training and validation assess unseen-center transfer; models including its training subset assess pooling with target-source examples. All five evaluated targets have supervision in each

Table 7 Measured transfer results on the fixed iBSAFE-MC test set. GB/CD/CBD/RS use box AP50; NGZ uses mask AP50–95 (%). B: combined iBSAFE-k/u; E: Endoscapes; I: iBSAFE-MC. The first two rows exclude iBSAFE-MC from training and validation (unseen-center transfer); the remaining rows include target-source training examples. Each row is one trained model.

Training sources	GB	CD	CBD	RS	NGZ
B	92.97	45.28	78.15	34.11	50.50
B + E	96.30	49.09	76.05	26.41	50.28
B + I	97.49	48.93	77.60	46.91	53.24
B + E + I	98.12	51.39	79.41	39.39	52.36
E + I	96.17	39.23	57.12	6.06	49.21

combination. Endoscapes contributes GB/CD masks and CBD boxes, while iBSAFE and iBSAFE-MC supply RS and NGZ annotations; iBSAFE-MC has no native line references.

For unseen-center transfer, adding Endoscapes to iBSAFE improves GB/CD but lowers CBD/RS and leaves NGZ slightly lower. Adding target-source examples to iBSAFE improves GB, CD, RS and NGZ, including a 12.80-point RS AP50 gain, while CBD decreases by 0.55 points. No source combination consistently outperforms the others across all anatomical targets, demonstrating that increased training-data diversity does not guarantee improved generalization. Importantly, performance on genuinely unseen surgical centers must be distinguished from performance obtained after incorporating training examples from those centers. Our experiments illustrate how training-data composition influences cross-center performance and why transparent reporting of dataset provenance is essential for interpreting generalization results.

4 Discussion

Our study demonstrates the feasibility of jointly learning safety-relevant anatomy from heterogeneous expert annotations, with generally improved CBD localization over single-target training. Beyond model performance, our experiments reveal important evaluation pitfalls. Differences between expert-selected and regularly sampled frames demonstrate the influence of test-set composition, while stratification by surgical phase and fluorescence evidence exposes limitations concealed by aggregate metrics. These findings motivate evaluating on common test populations, reporting sampling strategies, and stratifying results by clinically relevant conditions. Our cross-center experiments further highlight the importance of distinguishing generalization to unseen centers from evaluation after training on data originating from those centers.

Nevertheless, conventional detection and segmentation metrics do not fully capture clinical safety. In particular, NGZ underestimation and overestimation may have different consequences despite similar overlap scores. Following the problem-aware validation principles of *Metrics Reloaded* [9], future work should incorporate safety-specific error measures.

5 Conclusion

Joint learning from complementary expert annotations enables NGZ delineation and recognition of safety-relevant anatomy in laparoscopic cholecystectomy, with generally improved CBD localization over single-target supervision. Beyond model performance, our experiments demonstrate how frame sampling, surgical phase, local fluorescence evidence and training-data composition influence reported results and their interpretation. These findings highlight important evaluation pitfalls and provide practical recommendations for more reliable performance assessment and cross-center comparisons. Precise landmark recovery and CBD detection under weak fluorescence remain challenging, while temporal consistency and clinical safety require further validation.

Declarations

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References

[1] Carion N, Gustafson L, Hu YT, et al (2026) SAM 3: Segment Anything with Concepts. In: International Conference on Learning Representations

[2] Chen Q, Zhou R, Weng J, et al (2021) Extrahepatic biliary tract visualization using near-infrared fluorescence imaging with indocyanine green: optimization of dose and dosing time. Surgical Endoscopy 35(10):5573–5582

[3] Dip F, LoMenzo E, Sarotto L, et al (2019) Randomized Trial of Near-infrared Incisionless Fluorescent Cholangiography. Annals of Surgery 270(6):992–999

[4] Endo Y, Tokuyasu T, Mori Y, et al (2023) Impact of AI system on recognition for anatomical landmarks related to reducing bile duct injury during laparoscopic cholecystectomy. Surgical Endoscopy 37(7):5752–5759

[5] Gupta V, Jain G (2019) Safe laparoscopic cholecystectomy: Adoption of universal culture of safety in cholecystectomy. World Journal of Gastrointestinal Surgery 11(2):62–84

[6] Hu EJ, Shen Y, Wallis P, et al (2022) LoRA: Low-Rank Adaptation of Large Language Models. In: International Conference on Learning Representations

[7] Jenke AC, Bodenstedt S, Kolbinger FR, et al (2024) One model to use them all: training a segmentation model with complementary datasets. International Journal of Computer Assisted Radiology and Surgery 19(6):1233–1241

[8] Madani A, Namazi B, Altieri MS, et al (2022) Artificial Intelligence for Intraoperative Guidance: Using Semantic Segmentation to Identify Surgical Anatomy During Laparoscopic Cholecystectomy. Annals of Surgery 276(2):363–369

[9] Maier-Hein L, Reinke A, Godau P, et al (2024) Metrics reloaded: recommendations for image analysis validation. Nature Methods 21(2):195–212

[10] Mascagni P, Vardazaryan A, Alapatt D, et al (2022) Artificial Intelligence for Surgical Safety: Automatic Assessment of the Critical View of Safety in Laparoscopic Cholecystectomy Using Deep Learning. Annals of Surgery 275(5):955–961

[11] Mascagni P, Alapatt D, Murali A, et al (2025) Endoscapes, a critical view of safety and surgical scene segmentation dataset for laparoscopic cholecystectomy. Scientific Data 12(1):331

[12] Perazzi F, Pont-Tuset J, McWilliams B, et al (2016) A Benchmark Dataset and Evaluation Methodology for Video Object Segmentation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp 724–732

[13] Reinke A, Li ZO, Tizabi MD, et al (2026) Current validation practice undermines surgical AI development. arXiv preprint arXiv:2511.03769

[14] Smithmaitrie P, Khaonualsri M, Sae-Lim W, et al (2024) Development of deep learning framework for anatomical landmark detection and guided dissection line during laparoscopic cholecystectomy. Heliyon 10(3):e25210

[15] Sutherland F, Dixon E (2017) The importance of cognitive map placement in bile duct injuries. Canadian Journal of Surgery 60(6):424–425

[16] Tokuyasu T, Iwashita Y, Matsunobu Y, et al (2021) Development of an artificial intelligence system using deep learning to indicate anatomical landmarks during laparoscopic cholecystectomy. Surgical Endoscopy 35(4):1651–1658

[17] Yin SM, Lien JJJ, Chiu IM (2025) Deep learning implementation for extrahepatic bile duct detection during indocyanine green fluorescence-guided laparoscopic cholecystectomy: pilot study. BJS Open 9(2):zraf013